

OYEN AGDM, AB, Canada

WMO: 713420

Lat: 51.3797N

Lon: 110.3552W

Elev: 2515

StdP: 13.41

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.1	-16.1	-27.6	1.3	-20.8	-22.1	1.8	-16.1	29.6	23.4	26.2	21.9	8.4	250	0.736

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.6	87.9	62.6	84.0	61.4	80.2	60.4	66.3	80.1	64.3	77.6	62.5	75.5	10.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	91.6	70.9	59.6	83.7	69.1	57.5	77.5	66.6	32.4	80.1	30.8	77.6	29.3	75.9	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.5	24.0	21.0	-28.2	93.8	4.7	3.6	-31.6	96.4	-34.3	98.5	-36.9	100.6	-40.3	103.2		
			-28.0	70.1	4.9	2.9	-31.5	72.2	-34.4	73.9	-37.2	75.5	-40.8	77.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	13.6	12.2	26.0	39.8	51.5	59.1	65.0	62.7	53.8	40.6	26.0	14.0	
	DBStd	21.93	14.85	14.00	14.24	10.05	8.26	6.12	5.86	6.53	8.82	10.01	12.41	14.26	
	HDD50	5558	1127	1058	745	324	83	5	0	2	58	318	722	1116	
	HDD65	9702	1592	1478	1209	757	421	195	72	122	349	756	1170	1581	
	CDD50	1490	0	0	1	17	131	278	466	397	173	26	1	0	
	CDD65	161	0	0	0	0	4	19	73	52	13	0	0	0	
	CDH74	2622	0	0	1	5	132	306	1044	831	292	11	0	0	
	CDH80	854	0	0	0	0	22	76	375	282	99	0	0	0	
Wind	WSAvg	10.8	11.9	10.7	11.4	11.9	11.7	10.7	9.3	9.1	9.9	11.1	11.1	10.7	
Precipitation	PrecAvg	13.0	0.6	0.3	0.5	0.8	1.5	2.9	2.1	1.7	1.0	0.6	0.5	0.4	
	PrecMax	18.5	1.9	0.7	1.4	2.4	3.1	5.9	5.0	3.3	3.3	1.3	1.7	0.7	
	PrecMin	6.8	0.1	0.0	0.1	0.1	0.1	0.7	0.8	0.2	0.0	0.2	0.0	0.1	
	PrecStd	3.0	0.4	0.2	0.3	0.6	0.9	1.3	0.9	0.8	0.8	0.3	0.4	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	41.4	60.5	73.7	83.1	87.6	93.9	91.8	89.1	75.3	59.4	44.4	
		MCWB	36.7	36.2	46.6	51.2	57.4	63.1	65.4	61.7	59.7	54.7	46.2	38.3	
	2%	DB	39.4	37.6	53.5	67.0	77.6	81.6	88.4	86.9	82.1	67.5	50.4	40.1	
		MCWB	35.0	33.7	42.6	48.6	54.7	60.0	64.9	61.2	57.8	50.8	41.2	35.6	
	5%	DB	36.9	34.7	48.0	61.5	73.2	77.2	84.2	82.9	76.1	61.1	45.3	37.0	
		MCWB	33.5	31.6	40.0	45.4	53.0	59.3	63.0	61.0	55.9	47.5	38.1	33.0	
	10%	DB	33.8	31.4	42.8	56.6	68.7	73.1	79.8	78.5	70.5	55.8	40.6	32.8	
		MCWB	30.9	29.0	36.8	43.3	51.0	57.5	62.6	59.9	53.5	45.0	35.4	29.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	36.9	47.5	52.6	59.2	66.1	71.2	66.9	61.3	55.8	46.7	39.1
			MCDB	41.5	40.6	59.3	70.9	76.7	81.3	84.0	80.6	82.2	72.3	58.6	43.4
2%		WB	35.3	34.0	43.2	49.2	56.4	63.0	67.9	64.5	58.9	51.6	42.0	35.9	
		MCDB	38.7	37.1	51.8	64.5	73.6	75.8	81.2	79.3	78.6	64.8	49.3	39.9	
5%		WB	33.6	31.9	40.0	46.5	54.3	60.8	65.6	62.7	56.8	48.6	38.6	33.1	
		MCDB	36.8	34.7	47.2	59.6	69.8	73.8	78.9	77.1	73.9	59.3	44.0	36.7	
10%		WB	30.9	29.0	36.9	44.2	52.3	58.7	63.8	60.9	54.5	45.8	35.9	30.0	
		MCDB	33.7	31.5	42.3	54.7	66.2	70.2	76.3	74.9	68.7	55.0	40.1	32.7	
Mean Daily Temperature Range		5% DB	MDBR	17.2	17.0	18.4	21.7	23.7	21.3	24.6	24.7	23.7	21.0	16.6	15.7
			MCDBR	16.7	18.4	24.7	30.9	30.7	28.2	31.8	33.1	33.4	29.4	21.8	17.4
	MCWBR		13.7	15.0	15.0	15.6	13.9	12.2	12.7	12.7	14.1	15.4	14.2	13.8	
	5% WB	MCDBR	16.6	18.0	23.3	28.1	26.9	24.7	27.3	28.0	31.0	27.8	20.4	17.0	
		MCWBR	13.9	15.1	14.9	14.8	12.8	12.0	13.0	12.0	13.7	15.3	14.0	13.7	
Clear-Sky Solar Irradiance	taub	0.260	0.293	0.295	0.348	0.375	0.367	0.368	0.383	0.329	0.289	0.262	0.252		
	taud	2.354	2.308	2.436	2.301	2.266	2.348	2.376	2.296	2.434	2.519	2.473	2.363		
	Ebn at Noon	246	265	290	282	278	281	279	267	273	267	247	230		
	Edh at Noon	19	27	29	37	40	37	36	37	29	22	17	16		
All-Sky Solar Radiation	RadAvg	372	706	1139	1542	1857	1929	2075	1687	1213	735	390	280		
	RadStd	37	62	86	123	125	122	90	124	108	90	34	27		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page